Practical Application 1:

Overview

In this first practical application assignment of the program, you will seek to answer the question, "Will a customer accept the coupon?" The goal of this project is to use what you know about visualizations and probability distributions to distinguish between customers who accepted a driving coupon versus those who did not. Use the Practical Application 1 Jupyter Notebook Links to an external site.to complete this assignment. For this assignment, you will work with the Jupyter notebook locally. For more information on how this is done, please visit the Working with Jupyter Notebooks Locally page.

Data

This data is from the UCI Machine Learning Repository and was collected via a survey on Amazon Mechanical Turk. The survey describes different driving scenarios, including the destination, current time, weather, and passenger, and then asks people whether they will accept the coupon if they are the driver. There are three possible answers people can choose from:

- "Right away"
- "Later, before the coupon expires"
- "No, I do not want the coupon"

The first two responses are labeled as "Y = 1," and the third is labeled as "Y = 0." There are five different types of coupons: Less expensive restaurants (under $20), coffee houses, carryout and takeaway, bars, and more expensive restaurants ($20–$50).

Deliverables

Your final submission should include the Practical Application 1 Jupyter Notebook and other relevant files, such as the dataset provided and uploaded to a public-facing GitHub repository as your first portfolio project. To explore the data, you will utilize your knowledge of pandas and Python to create statistical summaries demonstrating differences in those who accepted and rejected the coupon. Utilize the Matplotlib and Seaborn libraries to create a visualization using Python. Ensure that your findings are clearly stated in a separate section alongside actionable items and recommendations.

Your repository should also include a READMe file containing a brief nontechnical report that highlights the differences between customers who did and did not accept the coupons.

Submission Instructions:

Submit the website URL to your public-facing GitHub repository here.

The average completion time for this assignment is 120 mins.

Required Assignment 5.1: Will the Customer Accept the Coupon?

Required Assignment 5.1: Will the Customer Accept the Coupon?

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeProject Organization · A READMe file with a summary of findings and a link to your Jupyter notebook · Jupyter notebook with headings and text appropriately formatted · No unnecessary files · Directories and files have appropriate names and locations	**5 pts** **Excellent** Your submission includes all listed components.	**0 pts** **Criterion Not Met** Your submission includes a few to of the listed components.	5 pt s

Required Assignment 5.1: Will the Customer Accept the Coupon?

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeSyntax and Code Quality · Libraries are imported and aliased correctly · Code does not contain errors · No long strings of code output · Demonstrates competency with pandas · Demonstrates competency with seaborn · Comments are used appropriately to explain code · Variables are sensible	**5 pts** **Excellent** Your submission includes all listed components.	**0 pts** **Criterion Not Met** Your submission includes a few tc of the listed components	5 pt s

Required Assignment 5.1: Will the Customer Accept the Coupon?

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeVisualizations · Appropriate plots for categorical and continuous variables are utilized · Plots contain human readable labels · Plots contain descriptive titles · Axes are legible · Subplots are used when appropriate · Plots are scaled appropriately for readability	**5 pts** **Excellent** Your submission includes all listed components.	**0 pts** **Criterion Not Met** Your submission includes a few to of the listed components.	5 pts

Required Assignment 5.1: Will the Customer Accept the Coupon?

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeFindings · Clearly stated problem for specific coupon group · Visualizations that demonstrate exploring differences in those who accepted and rejected the coupon · Interpretation of descriptive and inferential statistics is correct and concise · The findings are clearly stated in their own section with actionable items highlighted · Next steps and recommendations	**5 pts** **Excellent** Your submission includes all listed components.	**0 pts** **Criterion Not Met** Your submission includes a few to of the listed components.	5 pt s
Total Points: 20			

Practical Application 2:

Overview

In this application, you will explore a dataset from Kaggle. The original dataset contained information on 3 million used cars. The provided dataset contains information on 426K cars to ensure the speed of processing. Your goal is to understand what factors make a car more or less expensive. As a result of your analysis, you should provide clear recommendations to your client—a used car dealership—as to what consumers value in a used car.

To frame the task, throughout these practical applications, we will refer to a standard process in the industry for data projects called CRISP-DM. This process provides a framework for working through a data problem. Your first step in this application will be to read through a brief overview of CRISP-DM Links to an external site.**.**

Data

You will work with a used car dataset Links to an external site.for this assignment.

Deliverables

After understanding, preparing, and modeling your data, write up a basic report that details your primary findings. Your audience for this report is a group of used car dealers interested in fine-tuning their inventory.

Submission Instructions:

- Submit the website URL to your public-facing GitHub repository.
- Your Program Leader will grade your submission according to the rubric below.

The average completion time for this assignment is 15 hours.

Required Assignment 11.1: What Drives the Price of a Car?

Required Assignment 11.1: What Drives the Price of a Car?

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeProject organization: · A READMe file with a summary of findings and a link to the notebook · Jupyter notebook with headings and text appropriately formatted · No unnecessary files · Directories and files have appropriate names and locations.	**5 pts** **Excellent** Your submission includes all listed components.	**0 pts** **Criterion not met** Your submission includes a few none of the listed components.	5 pts

Required Assignment 11.1: What Drives the Price of a Car?

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeSyntax and code quality: · Libraries are imported and aliased correctly. · Code does not contain errors. · No long strings of code output · Demonstrates competency with pandas · Demonstrates competency with seaborn · Comments are used appropriately to explain code. · Variables are sensible.	**5 pts** **Excellent** Your submission includes all listed components.	**0 pts** **Criterion not met** Your submission includes a few none of the listed components.	5 pts

Required Assignment 11.1: What Drives the Price of a Car?

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeVisualizations: · Appropriate plots for categorical and continuous variables are utilized. · Plots contain readable labels. · Plots contain descriptive titles. · Axes are legible. · Subplots are used when appropriate. · Plots are scaled appropriately for readability.	**5 pts** **Excellent** Your submission includes all listed components.	**0 pts** **Criterion not met** Your submission includes a few none of the listed components.	5 pts

Required Assignment 11.1: What Drives the Price of a Car?

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeModeling: · Use of multiple regression models · Cross-validation of models · Grid search hyperparameters · Appropriate interpretation of coefficients in models · Appropriate interpretation of evaluation metric · Clear identification of evaluation metric · Clear rationale for use of given evaluation metric	**5 pts** **Excellent** Your submission includes all listed components.	**0 pts** **Criterion not met** Your submission includes a few none of the listed components.	5 pts

Required Assignment 11.1: What Drives the Price of a Car?

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeFindings: · Clearly stated business understanding of the problem · Clean and organized notebook with data cleaning · Correct and concise interpretation of descriptive and inferential statistics · Clearly stated findings in their own section with actionable items highlighted in an appropriate language for a nontechnical audience · Next steps and recommendations	**5 pts** **Excellent** Your submission includes all listed components.	**0 pts** **Criterion not met** Your submission includes a few none of the listed components.	5 pts
Total Points: 25			

Practical Application 3:

Overview

In this third practical application assignment, your goal is to compare the performance of the classifiers (k-nearest neighbors, logistic regression, decision trees, and support vector machines) you encountered in this section of the program. You will use a dataset related to the marketing of bank products over the telephone.

Data

The dataset you will use comes from the UC Irvine Machine Learning Repository Links to an external site.. The data is from a Portuguese banking institution and is a collection of the results of multiple marketing campaigns. Please see the following link to access the assignment:

Required Assignment 17.1Links to an external site.

Deliverables

After understanding, preparing, and modeling your data, build a Jupyter Notebook that includes a clear statement demonstrating your understanding of the business problem, a correct and concise interpretation of descriptive and inferential statistics, your findings (including actionable insights), and next steps and recommendations.

Submission Instructions:

- Submit the website URL to your public-facing GitHub repository here.
- Your Program Leader will grade your submission according to the rubric below.

The average completion time for this assignment is 15 hours.

Required Assignment 17.1: Comparing Classifiers

Required Assignment 17.1: Comparing Classifiers

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeProject organization: • A READMe file with a summary of findings and a link to the Jupyter Notebook. • Jupyter Notebook has headings and text appropriately formatted. • No unnecessary files. • Directories and files have appropriate names and locations.	**5 pts** **Excellent** Your submission includes all of the listed components.	**0 pts** **Criterion not met** Your submission includes a fe none of the listed component	5 pt s

Required Assignment 17.1: Comparing Classifiers

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeSyntax and code quality: • Libraries are imported and aliased correctly. • Code does not contain errors. • No long strings of code output. • Demonstrates competency with pandas. • Demonstrates competency with seaborn. • Comments are used appropriately to explain code. • Variables are sensible.	**5 pts** **Excellent** Your submission includes all of the listed components.	**0 pts** **Criterion not met** Your submission includes a fe none of the listed component	5 pt s

Required Assignment 17.1: Comparing Classifiers

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeVisualizations: • Appropriate plots for categorical and continuous variables are utilized. • Plots contain readable labels. • Plots contain descriptive titles. • Axes are legible. • Subplots are used when appropriate. • Plots are scaled appropriately for readability.	**5 pts** **Excellent** Your submission includes all of the listed components.	**0 pts** **Criterion not met** Your submission includes a fe none of the listed component	5 pts

Required Assignment 17.1: Comparing Classifiers

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeModeling: • Use of multiple machine learning models. • Cross-validation of models. • Grid search hyperparameters. • Appropriate interpretation of coefficients in models. • Appropriate interpretation of an evaluation metric. • Clear identification of an evaluation metric. • Clear rationale for use of the given evaluation metric.	**5 pts** **Excellent** Your submission includes all of the listed components.	**0 pts** **Criterion not met** Your submission includes a fe none of the listed component	5 pt s

Required Assignment 17.1: Comparing Classifiers

Criteria	Ratings		Pts
This criterion is linked to a Learning OutcomeFindings: • Clearly stated business understanding of the problem. • Clean and organized notebook with data cleaning. • Correct and concise interpretation of descriptive and inferential statistics. • Clearly stated findings in their own section with actionable items highlighted in appropriate language for a nontechnical audience. • Next steps and recommendations.	**5 pts** **Excellent** Your submission includes all of the listed components.	**0 pts** **Criterion not met** Your submission includes a fe none of the listed component	5 pt s
Total Points: 25			